GNBF5020

Midterm Exam Focus Prediction

What Dr. Tony Wong will actually test, and why

GNBF5020 · Introduction to Molecular Biology and Genetics

Dr. Tony Wong · Lectures 1-5, 12 Sep - 17 Oct 2026

[ENGLISH ONLY] **THE EXAM MUST BE ANSWERED IN ENGLISH.** Dr. Wong said this three separate times, and once he said he gave zero marks for it. Every assignment and every exam in this course must be written in English.

[HOW TO READ THIS] Derived from a second, targeted listening pass over all five lecture recordings (25 audio chunks) plus every slide deck (~454 frames). This is a prediction from teaching behaviour - not a leaked paper.

Video guides

Twelve terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index.html** plays all twelve on one page.

Video	Term	Chapter	Video file
V01	**Kaplan–Meier curve**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell**	Ch 8 · Cell biology	_video/assets/V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID**	Ch 6 · DNA damage and repair	_video/assets/V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree**	Ch 8 · Cell biology	_video/assets/V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)**	Ch 8 · Cell biology	_video/assets/V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER**	Ch 6 · DNA damage and repair	_video/assets/V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour**	Ch 9 · Immunology and cancer immunity	_video/assets/V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4**	Ch 9 · Immunology and cancer immunity	_video/assets/V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker**	Ch 11 · Assay techniques and omics	_video/assets/V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans**	Ch 7 · Control of gene expression	_video/assets/V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation**	Ch 4 · Chromatin and epigenetics	_video/assets/V12_de_novo_vs_maintenance_methylation.mp4

Open the video index page ->

[!] Read this first - the exam must be answered in English

Dr. Wong said this three separate times, and once he said he gave zero marks for it:

> *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong."* - Lecture 1
>
> *"I'm not kidding, because last year there are some student they answered the question in Chinese, and I have no choice that I give the zero mark, because everything should be in English."* - Lecture 1
>
> *"you have to answer in English, although this is a CUHK, haha, okay."* - Lecture 4

This document is in English so you can revise directly in the language you will be graded in. The Chinese companion (`GNBF5020_考試重點預測_繁體中文.pdf`) exists only for explanation; **gene, protein, pathway, method and drug names stay in English in both versions.**

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1. The exam, exactly as he describes it

Item	Specification	Source
Weight	**Midterm = 30%** of final grade; final exam = 40%; two "super-easy" assignments = 30%	Lecture 1, Slide 2
Part A	**Multiple choice - single best option only.** Last year students lost marks by trying to select multiple options	Lecture 4, Lecture 5
Part B	**Data-analysis / scenario questions.** Given a set, choose one to answer	Lecture 4
Language	**English only - zero marks for Chinese answers**	Lectures 1, 4, 5
Duration	**1.5 hours (90 minutes)**	Lecture 5

[!] The two format statements do not match - use the later one

[FORMAT CONFLICT] Lecture 4 (10 Oct 2026) says 25 MCQ and choose 1 of 2 short questions; Lecture 5 (17 Oct 2026, later and written on his own title slide) says 20 MCQ and choose 2 of 3 short questions. Plan for Lecture 5 - but the ratio and the strategy are the same either way.

Lecture	Date	What he says
Lecture 4	10 Oct 2026	"about 25 questions, one mark each ... 25% of the final grade ... two questions, you choose only one short question to answer"
Lecture 5	**17 Oct 2026**	**"we have 20 multiple choice questions, and two out of three short question ... 20 marks ... short question is account for 10 marks ... 1.5 hours"**

Lecture 5 is the most recent statement and the one written on the title slide in his own handwriting, so plan for:

[EXAM FORMAT] 20 MCQ (20 marks) + choose 2 of 3 short questions (10 marks) = 30 marks, 90 minutes

Either way the ratio is the same and the strategy is the same: **MCQs are for passing; the scenario questions are where the marks are.**

> *"multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you'd better perform well for the short questions, which emphasize on data analysis and also the scenarios questions."* - Lecture 1

The two scenario-question archetypes he named outright

> *"For example, I will ask: if you're a genetic consultant and I give you a medical report, how you would suggest the client whether she or him will get a baby with a certain genetic diseases? Or... I give you some survival data, and then you tell me that how this survival data can really benefit the treatment of the patient."* - Lecture 1

So the short questions come from **two families only**: **a pedigree/genetic-counselling report** and **a survival curve / dataset**. Practise those two, not everything.

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2. How this prediction was built

Every 25-minute chunk of all five lectures was re-analysed with a prompt aimed at exam evidence. Four signal types were collected and ranked:

Signal	Strength	Meaning
He names the question as exam-type	**A+**	He says "this is the kind of question I would include in the exam"
He says "you must 100% understand / remember / always emphasize"	**A**	Explicitly required knowledge
He says "this is NOT in the exam"	**Negative evidence**	Explicitly excluded - saves you time
He labours over it with precise numbers	**B**	Standard MCQ fodder

A note on method, so you can weight this properly: this is a **prediction from teaching behaviour, not a leaked paper**. Tiers A+ and A rest on his own words, so they are as reliable as evidence gets. Tier B rests on pattern recognition, so treat it as strong guidance rather than certainty.

Figure E1 - Evidence-ranked focus map

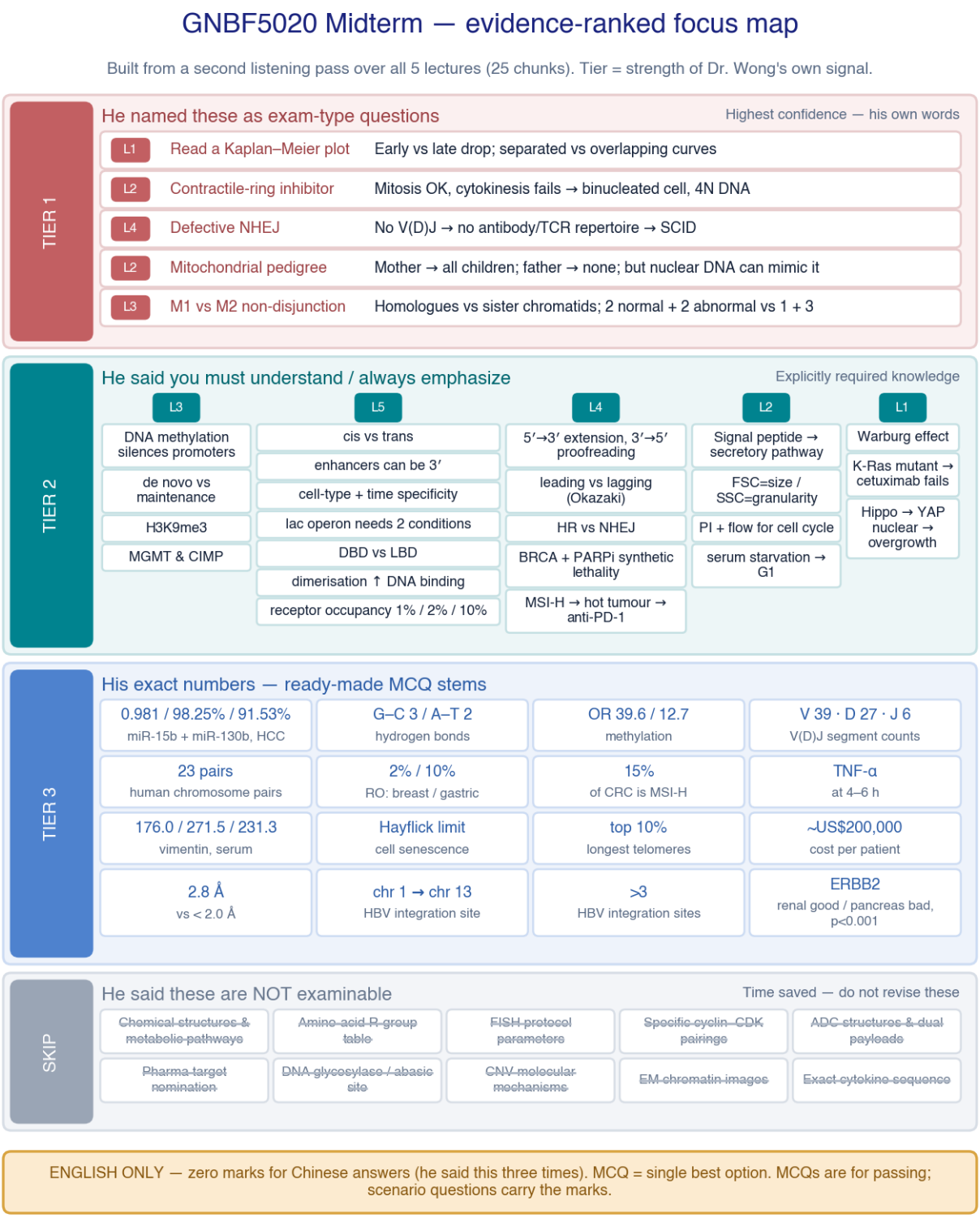


Figure E1. Tier 1 (his own words), Tier 2 (must understand), Tier 3 (his exact numbers) and the skip list, ranked by the strength of Dr. Wong's own signal. Figure authored at 6.5 in print width and placed 1:1 on this page.

3. TIER 1 (A+) - Questions he explicitly flagged as exam-type

3.1 Reading a Kaplan-Meier curve

LECTURE 1

> *"And I ask this single question every, every time when I talk about the KM plot, yeah. And only a single student give me a good answer last year, yeah."*
>
> *"So again, this is the kind of the question that I would include in the exam, yeah. Just give you a taste about the midterm exam, yeah."*
>
> *"But the key is to let you understand that from a single plot, we can really, really find out many, many useful informations for the future study."* - Lecture 1

The core skill: from one KM plot, deduce *early vs late* involvement of a gene.

- x-axis = months; y-axis = probability of survival (or progression-free survival)
- **Curve starts at 1.0 and falls in steps**; flat segments = no events in that interval; **downward steps = events**; small tick marks on the curve = **censored patients**
- **Curve that drops steeply and early** = the gene is associated with **early** involvement / poor prognosis
- **Curve that drops late** = **late** involvement
- **Two curves separate** = the gene separates the two patient groups
- **Curves that cross or overlap** = the gene does **not** separate prognosis

The lecture's worked example: the same target is *pro-tumour in one cancer and anti-tumour in another* - a receptor tyrosine kinase predicts poor survival in pancreatic cancer but **better** survival in renal cell carcinoma. ERBB2 in the Human Protein Atlas is the parallel: prognostic in renal cancer (**favourable**), endometrial cancer (**unfavourable**) and pancreatic cancer (**unfavourable**), all at **p < 0.001**.

Why this matters: he says this question is asked **every single time** he teaches KM, and only one student got it right last year. It is also literally one of the two scenario-question archetypes he named.

3.2 Contractile-ring inhibition -> binucleated cell

LECTURE 2

> *"Again, like I - I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly, yeah... Actually quite a surprise to me, yeah. So again this is the type of the question that I would try to include in the midterm, yeah. So when you do the revision, try to think in this way, yeah."* - Lecture 2

The question: you treat dividing cells with an inhibitor of the contractile ring (or knock out its expression). What do you see under the microscope / on a flow-cytometry DNA-content plot?

The answer:

- The contractile ring (actin + myosin) drives **cytokinesis** - the *cytoplasmic* division
- Mitosis (chromosome separation) is **independent** of cytokinesis and proceeds normally
- Genetic material replicates and divides, but the cell **cannot cleave into two daughter cells**
- Result: **one cell with two nuclei (a binucleated cell), with 4N DNA content** - flow cytometry shows a 4N peak instead of separate 2N and 4N peaks
- **Mitosis vs cytokinesis are separate processes** - this distinction is the whole point of the question

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3.3 NHEJ defect -> SCID

LECTURE 4

> *"Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen. Yeah, that's the clinical outcome of this genetic defect."* - Lecture 4

The chain you must be able to draw:

NHEJ is co-opted by V(D)J recombination to assemble V + D + J segments -> generates antibody (B-cell) and TCR (T-cell) diversity -> **defect in NHEJ -> cannot build the repertoire -> SCID (Severe Combined Immunodeficiency)**

Segment copy numbers he gave: **V ~ 39 copies, D = 27 copies, J = 6 copies.**

3.4 Mitochondrial inheritance and the pedigree LECTURE 2

> *"So, this is you'll have to a 100% understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA."*
>
> *"Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree, yeah."* - Lecture 2

This is a pedigree-reading question and therefore a scenario question. The three hallmarks of maternal (mitochondrial) inheritance:

- An affected **mother** passes the condition to **all** of her children
- An affected **father** passes it to **none** of his children (sperm mitochondria are destroyed after fertilisation)
- The condition appears in **every generation**

The twist that catches people: mitochondrial disease is **not** always mtDNA. Mutations in **nuclear** genes encoding mitochondrial proteins cause the same phenotype but with a **completely different inheritance pattern** (Mendelian, not maternal). A pedigree of "mitochondrial-looking" disease can therefore be nuclear. The answer must mention both.

The threshold effect (heteroplasmy): the presence of mutant mtDNA does not guarantee disease. Disease appears only when the **proportion of mutant mtDNA exceeds a tissue-specific threshold**. Tissues with high energy demand (e.g. contracting **skeletal muscle**) have a **low** threshold and are affected first; low-demand tissues (e.g. **hearing**) have a **high** threshold.

3.5 Meiosis I vs Meiosis II non-disjunction LECTURE 3

> *"And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1..."* - Lecture 3
>
> *"And I think you need to distinguish these two situations here, yeah."* - Lecture 3

	Meiosis I (M1)	Meiosis II (M2)
What fails to separate	**Homologous** chromosomes	**Sister chromatids**
Gametes from one meiosis	**2 normal + 2 abnormal** (all abnormal are the same type)	**1 normal + 3 abnormal** (2 of one type, 1 of another)
Recombination (crossover)	Occurs between homologues	Does not occur

Related - inversions: an inversion changes **no** genetic material and **no** gene dosage, yet it can drastically change expression by moving a gene away from its enhancer (or placing an enhancer next to a different promoter). *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."*

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4. TIER 2 (A) - Concepts he said you must understand / remember

4.1 DNA methylation and epigenetics LECTURE 3

> *"The most important aspect of the DNA methylation is it has a very significant functions on these gene expressions, because DNA methylation is usually involved in the suppression of the gene expressions."* - Lecture 3

- CpG methylation -> **suppresses** transcription; **promoter hypermethylation of tumour-suppressor genes** is a common cancer driver (silences the brake)
- **De novo** methyltransferases establish new marks; **maintenance** methyltransferases copy marks onto daughter strands after replication
- **Histone code nomenclature:** `H3K9me3` = Histone 3, Lysine at position 9, trimethylation
- **Cross-talk:** DNA methylation histone marks reinforce each other in **positive feedback loops** -> cellular memory
- **Clinical methylation biomarkers: MGMT** promoter methylation (glioblastoma -> sensitises to alkylating chemo + radiotherapy, better survival) and **CIMP** (colorectal, poorer outcome)
- **Odds Ratio ~ 12.7** for detected methylation; CIMP-high vs CIMP-0 **OR 39.6 (95 % CI 20.6-76.1), p < 0.0001**
- **Bisulfite sequencing** = gold standard (unmethylated C -> U; methylated C preserved)

4.2 Cis vs trans regulation LECTURE 5

- **Cis** elements (promoter, enhancer, silencer, insulator) act on **the same DNA molecule** they sit on
- **Trans** factors (transcription factors, co-activators) are diffusible and can act across chromosomes - he notes trans-regulation is *"much more common at least in eukaryotic system"*
- His example: a repressor encoded on **chromosome 1** acting on a target gene on **chromosome 13**
- **Not all enhancers are upstream.** *"Not all the enhancer are located on the upstream of the sequence. Some of the regulatory sequences can be located in the downstream sequence, so that means in the 3' ends of your target genes. This is one of the concepts that you need to understand."*
- **Cell-type specific AND time specific:** *"the expression is usually cell-specific and at the same time, it is time-specific... This is the concept that I always emphasize."*

4.3 The lac operon - two conditions, not one

LECTURE 5

Both must be true for strong transcription:

- **Lactose present** -> allolactose binds the **Lac repressor** -> it can no longer block the operator
- **Glucose low** -> cAMP high -> **CAP-cAMP** binds the promoter -> recruits RNA polymerase

`trp` operon is the mirror image: **tryptophan binds the Trp repressor** (corepressor) -> shuts down tryptophan biosynthesis.

- **IPTG** is the artificial, non-metabolised inducer used in *E. coli* recombinant systems
- **CHO cells** (mammalian) are used when glycosylation is required - and require **codon optimisation**: *"So remember that when you use the CHO cell, you need to do the codon optimization."*

4.4 "Hot" vs "cold" tumours

LECTURE 5

> *"So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor."* - Lecture 5

- **MSI-H / MMR-deficient -> frameshift mutations -> high neoantigen burden -> "hot tumour" -> responds to anti-PD-1**
- Cold tumours can be re-programmed hot (e.g. HDAC inhibitor inducing PD-L1 -> **"gene addiction"** -> checkpoint blockade)

  "I would say this figure is one of the most important figures in modern medicine, because when you attend any seminars about the immune checkpoint inhibitors or immunotherapies or colorectal cancer, I can for sure that the presenter will include this figure in his presentation deck." - Lecture 4 (the dMMR vs pMMR anti-PD-1 KM curve)

4.5 DNA replication directionality

LECTURE 4

> *"So this is important, you need to understand the elongation happens from 5 prime to 3 prime."*
>
> *"I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one is the lagging strand."* - Lecture 4

	Direction
Nucleotide addition (elongation)	**5' -> 3'**
Proofreading / excision	**3' -> 5'** (exonuclease activity)
Ligase seals	5' phosphate -> 3' hydroxyl (phosphodiester bond)

- **Leading strand**: continuous, one primer. **Lagging strand**: discontinuous **Okazaki fragments**, each needs its own RNA primer
- Replication bubbles: **thousands of origins**, each with **two bidirectional forks**
- **Proofreading loss** -> high neoantigen burden -> immunotherapy-responsive; also excises nucleoside-analogue drugs -> drug resistance

4.6 HR vs NHEJ - the single most-likely MCQ pair LECTURE 4

	HR	NHEJ
Template	**Yes** - sister chromatid	**No**
Fidelity	**Error-free**	**Error-prone** (indels/frameshifts)
Cell cycle	**S/G2 only**	All phases
Used by	Meiotic recombination	**V(D)J recombination -> SCID if defective**; **CRISPR knock-out**
CRISPR use	**Knock-in** (needs donor template)	**Knock-out**

4.7 Synthetic lethality and PARP inhibitors LECTURE 4

- BRCA1/2 loss -> HR deficient -> **PARP inhibitor** kills the tumour while normal cells survive on intact HR -> **wide therapeutic window**
- **Therapeutic window** = margin between minimum effective dose (efficacy) and toxic dose (toxicity). Synthetic lethality widens it.
- **CRISPR knockout vs old siRNA/shRNA** - CRISPR has largely replaced knockdown. (*"CRISPR/Cas9 is now standard for gene knock-out/knock-in; older siRNA/shRNA knockdown techniques are largely superseded."* - Lecture 2)

4.8 Nuclear receptor domains, dimerisation and receptor occupancy LECTURE 5

- **NTD** (variable, regulatory) / **DBD** (conserved, binds DNA response elements) / **LBD** (binds ligand -> conformational change -> recruits co-activators or co-repressors)
- **Monomers bind DNA poorly. Dimerisation dramatically increases affinity.** HER2 **cannot** homodimerise efficiently - it heterodimerises with HER1/HER3/HER4; **Herceptin blocks this dimerisation**
- **Tamoxifen (SERM):** antagonist in breast (anti-tumour) / partial agonist in endometrium -> small increased endometrial cancer risk. *"This is the concept that you have to bear in mind."*
- **Receptor occupancy (RO)** thresholds he stated: typical antibody **1%**; **Herceptin in breast cancer 2%**; **Herceptin in gastric cancer 10%** -> same drug, different dosing, because the required RO is tissue-specific

4.9 Signal peptide LECTURE 2

> *"So remember, signal peptide is really, really important for your study, yeah"* (after his postdoc lost **4 months** of work by omitting it)

- N-terminal signal peptide routes the nascent protein into the secretory pathway: rough ER -> Golgi -> plasma membrane / secretion
- **Removing it blocks expression of secreted and membrane proteins entirely**
- **Not** to be confused with the **N-terminal domain (NTD)** of a nuclear receptor (Lecture 5)

4.10 Chromatin accessibility

LECTURE 3

- **Euchromatin** = open, transcriptionally active; **Heterochromatin** = condensed, silent - it physically blocks transcription factors and enhancers from reaching the promoter
- Nucleosomes slide along DNA to expose or cover promoters; two drivers: **intrinsic DNA-histone affinity** and **ATP-dependent chromatin remodelling complexes**
- Histone-DNA binding is **non-covalent** (H-bonds + electrostatic) so it can be dynamic and reversible
- Histones are **highly conserved** across species (Drosophila -> human), which is why model organisms work

4.11 Karyotyping resolution

LECTURE 3

- G-banding detects gross structural change at roughly **0.5-10 Mb** (he quotes 5-10 Mb in one chunk, 0.5-1 Mb in another - both are in his material)
- **Point mutations / single-nucleotide variants require sequencing** (Sanger or NGS) - karyotyping misses them
- **p arm** = short, **q arm** = long, **centromere** = spindle attachment. Pronounced *"q-two-one-point-one"*, **not** "q-twenty-one-point-one"

4.12 Driver vs passenger genes

LECTURE 3

- **Driver** = confers a growth advantage. **Passenger** = hitching a ride, carried along by chromosomal proximity or instability
- To find true drivers you must **integrate CNV profiles with mRNA expression** - concordant copy gain + transcriptional upregulation. Worked example: **NPM1** in leukaemia

4.13 Flow cytometry fundamentals

LECTURE 2

- **FSC = size**, **SSC = granularity/internal complexity**
- **Gating** isolates the target population; **FACS** physically sorts viable cells into tubes/wells
- **PI staining** quantifies DNA content -> G1 (2N) / S / G2-M (4N)
- **Serum starvation** (FBS down to 0.5-1%) arrests cells at G1 to synchronise them
- **IHC vs IF:** IHC = chromogenic enzyme substrate; IF = fluorescent dye
- **FISH = DNA probes, not antibodies** -> counts gene copies -> HER2 amplification -> Herceptin candidacy
- **LCM** cuts a homogeneous tissue region -> avoids stromal/immune contamination in tumour biopsies
- Stem-cell QC he flagged: *"unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it."*

5. TIER 3 (B) - Precise numbers he stated: ready-made MCQ stems

Value	Context	Lecture
G-C = 3 hydrogen bonds; A-T = 2	GC-rich DNA needs higher Tm to denature -> PCR primer design	1, 3
GC content -> melting temperature: more GC means a higher Tm (his slide states the direction; the exam notes also carry a "~40 deg C drop" and a "60 deg C vs 55 deg C" board example, and the two do not reconcile)	GC content -> melting temperature	3
2.8 A nucleosome crystal structure; **< 2.0 A** = "very excellent"	X-ray resolution	3
23 pairs of chromosomes; **Trisomy 21**; **Turner 45,X**	Aneuploidy	3
~10 nm nucleosome diameter	"Beads on a string"	3
Telomere 3-20 kb TTAGGG; **100-300 kb** subtelomeric	Chromosome end structure	3
V ~ 39, D = 27, J = 6	V(D)J gene segment copies	4
Serum starvation to 0.5-1 % FBS	G1 arrest / synchronisation	2, 4
OR 39.6 (CIMP-high vs CIMP-0, 95 % CI 20.6-76.1, p < 0.0001)	CDKN2A methylation	3
OR 12.7 (7.84-20.7, p < 0.0001)	CDKN2A methylation / p16 loss	3
miR-15b + miR-130b: AUC 0.981, sensitivity **98.25 %**, specificity **91.53 %**, accuracy **94.83 %**	Circulating miRNA classifier for HCC	1
Serum vimentin: controls **176.0 +/- 61.3**; HCC <= 2 cm **271.5 +/- 165.9**; HCC > 2 cm **231.3 +/- 132.3** ng/mL	HCC biomarker	1
15 % of colorectal cancers are MSI-H	MMR deficiency	4
~US$200,000 per patient; **Phase 1A** dose escalation, **1B** expansion, **2/3** efficacy	Clinical trial design	2, 4
Receptor occupancy: 1 % / 2 % (breast) / 10 % (gastric)	RO thresholds	5
TNF-alpha peaks at 4-6 h, then IL-1beta -> IFN-gamma -> IL-2 / IL-15	NF-kappaB cytokine time course	5
Chromosome 1 repressor -> chromosome 13 target	cis vs trans example	5
3 or more HBV integration sites -> significantly poorer survival	HBV integration as prognostic marker	2
Top 10 % longest telomeres -> worse recurrence-free survival	Telomere length in HCC	4
ERBB2 prognostic in renal (favourable), endometrial (unfavourable), pancreatic (unfavourable), all p < 0.001	Human Protein Atlas	1
Rasonque (daraxonrasib) median OS **13.2 vs 6.7 months**	K-Ras G12C inhibitor	1
Anti-TERT antisense oligonucleotide: 3 weeks pretreatment -> tumour take 2/12 (vs 9/12 untreated), p = 0.01	HCC tumourigenesis	4

6. What he explicitly said is NOT examinable - skip it

SKIP LIST — he put these out of scope, so do not revise them

Not examinable	His words	Lect
~~Chemical structures, metabolic pathway details~~	*"Please don't memorize all these chemical structure... They would not be included in my exam."*	1

~~The amino acid R-group table~~	*"don't memorize this... definitely not include in the exam"*	1
~~FISH protocol parameters (denaturation, temperature)~~	*"Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature"*	2
~~Specific cyclin-CDK pairings~~	*"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK"*	2
~~Detailed signalling cascades on the slide~~	*"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology"*	2
~~**ADC structures, dual-payload ADCs**~~	*"all this would not be included in the midterm. I'm just want you to know more"*	4
~~Pharma target nomination / druggability workflow~~	*"it's not included in exam, but I think it's really more interesting"*	4
~~**DNA glycosylase and abasic (AP) site chemistry**~~	*"I can ensure that they will not be included in the coming midterm"*	4
~~CNV molecular mechanisms in depth~~	*"I'm just give you a taste... don't be scared with all these mechanism"*	3
~~Electron microscopy chromatin images~~	*"don't memorize all these figures, yeah. It's not necessary"*	3
~~The exact cytokine sequence~~	*"don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times."*	5

7. The pairs most likely to be used as MCQ distractors

These are the classic "which one is different?" structures. He repeatedly returns to each pair, and each pair has exactly one discriminating fact.

Pair	Discriminator
BER vs NER	BER = single damaged base; NER = bulky multi-nucleotide lesions (e.g. UV dimers)
MMR vs HR	MMR = replication mismatches + small indels; HR = high-fidelity DSB repair
HR vs NHEJ	HR uses a template, S/G2 only; NHEJ no template, all phases, error-prone
cis vs trans	cis = same DNA molecule; trans = diffusible factor encoded elsewhere
IgG1 vs IgG4	IgG1 = ADCC; IgG4 = no ADCC, longer half-life
Autologous vs allogeneic CAR-T	patient-derived vs donor/iPSC-derived
In vivo CAR-T	mRNA in lipid nanoparticles, engineered in the body
Driver vs passenger	causal growth advantage vs bystander
Euchromatin vs heterochromatin	active vs condensed/silent
DBD vs LBD	DNA-binding vs ligand-binding domain
de novo vs maintenance methylation	establish new mark vs copy onto daughter strand
Predictive vs prognostic biomarker	predicts *response to therapy* vs predicts *disease outcome*
Dimerisation vs monomer	monomers bind DNA poorly; dimers bind strongly
5'->3' vs 3'->5'	elongation vs proofreading
M1 vs M2 non-disjunction	homologues vs sister chromatids

Pair	Discriminator
mtDNA vs nuclear DNA	maternal inheritance vs Mendelian, different threshold logic
ADCC vs CDC	antibody-dependent cellular vs complement-dependent cytotoxicity
Signal peptide vs NTD	ER targeting vs nuclear receptor regulatory domain
Exon 1 vs Exon 2 methylation	Exon 2 is the diagnostically informative region

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8. Likely MCQ stems

1	*"Which pair of statements about X and Y is correct?"* -> the pairs in 7
2	*"Which enzyme/pathway is responsible for X?"* -> repair pathway matching; or the cascade order (LPS -> TLR4 -> IKK -> IkappaB -> NF-kappaB -> nucleus)
3	*"A patient with X is most likely to respond to which drug?"* -> K-Ras mutant -> cetuximab fails; BRCA-mutant -> PARPi; MSI-H -> anti-PD-1; HER2+ -> trastuzumab; promoter hypermethylation -> HDAC inhibitor
4	*"According to the Kaplan-Meier plot, which group has better/worse prognosis?"* -> early vs late drop; separation vs overlap
5	*"What is the consequence of inactivating X?"* -> NHEJ -> SCID; contractile ring -> binucleated cell; telomerase -> senescence; mismatch repair -> MSI-H
6	*"Which value is correct?"* -> the numbers in 5
7	*"Which is NOT a function of X?"* -> the negative-evidence items in 6
8	*"A child's pedigree suggests mitochondrial disease - explain the inheritance pattern."* -> 3 hallmarks + the nuclear-DNA caveat + threshold effect

9. Likely short-question (scenario) archetypes

These are the archetypes from the existing 16-case bank in `GNBF5020_Case_Studies.md`, re-ranked by this document's evidence. Cases 1-6 and 10-13 are the highest-yield.

No.	Scenario	What the marker wants
1	**KM plot** -> read it	Early vs late drop; separation vs overlap; which gene; which cancer type
2	**Pedigree + risk** -> genetic counselling	Inheritance pattern; carrier probability; why mutation != certainty (incomplete penetrance, expression regulation)
3	**Survival data** -> treatment benefit	Read the KM/multivariate result; link to the mechanism; state the clinical action
4	**Flow plot**	FSC vs SSC; which gate; which population; what the contaminant is
5	**Binucleated cells after treatment**	Mitosis != cytokinesis; contractile ring; 4N DNA content
6	**Companion diagnostic**	Which test (FISH/IHC/IF/ELISA/LCM), what it measures, when it is used
7	**MGMT methylation result**	Predicts alkylating-chemo + RT sensitivity; mechanism
8	**Pedigree + aneuploidy**	M1 vs M2; resulting gametes; syndrome
9	**Inversion without dosage change**	Enhancer-promoter repositioning; expression changes without DNA gain/loss
10	**Tumour resists the obvious drug**	Secondary resistance; synthetic lethality; PARPi in BRCA
11	**MMR / MSI result**	MSI-H -> hot tumour -> anti-PD-1; CEA as surrogate marker
12	**Choosing an editing strategy**	Knock-out (NHEJ) vs knock-in (HR, S/G2); knockdown vs knockout

No.	Scenario	What the marker wants
13	**Why is a topoisomerase inhibitor in the antibody?**	ADC rationale; selective delivery; payload
14	**Compound that switches itself off in the right cells**	HRE-controlled CAR-T; hypoxia; one gate
15	**Isoform that predicts drug resistance**	Alternative splicing; splice site creation; neoantigen
16	**Reading the lac operon logic correctly**	Two conditions; allolactose; cAMP-CAP; IPTG

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10. English answer templates for the scenario questions

Dr. Wong said he is *"emphasising on how you analyze the scientific questions"*, not on memorised facts. So a short question is won by **structure**, not volume. Every answer should have four moves:

1	**State the observation** - what the data shows, in one sentence, with the number
2	**Name the mechanism** - the pathway/process responsible
3	**Interpret the clinical meaning** - what it means for the patient
4	**State the action or the limit** - what you would do, or what you cannot conclude

Skeleton (use this for every short question):

```
"From the [plot/table/pedigree], [observation with the specific value]. This is consistent with [mechanism], because [reason]. Clinically this means [interpretation for the patient]. Therefore I would [action]. Note that we cannot conclude [limitation], because [reason]."
```

KM-plot template:

```
"The curve for [group A] drops from 1.0 to [x] within [n] months, whereas [group B] remains near [y] until [m] months. The [log-rank] p-value is [p], so the two groups are separated. A steeper and earlier drop indicates [early involvement / worse prognosis]. This is consistent with [mechanism]. The same target can behave in opposite directions in different tumour types - [ERBB2 example] - so this finding should not be generalised beyond [this tumour type]. The practical implication is [action]."
```

Pedigree template:

```
"The pedigree shows [transmission pattern], which is consistent with [mitochondrial / autosomal dominant / X-linked] inheritance. The three features supporting this are [feature 1], [feature 2], [feature 3]. Importantly, [mitochondrial disease can also arise from nuclear gene mutations], so the pedigree cannot be assigned to mtDNA on pattern alone; confirmation would require [sequencing both the mitochondrial genome and relevant nuclear genes]. The recurrence risk for the proband is [x]%. This is a probability, not a certainty, because of [incomplete penetrance / threshold effect / variable expression]."
```

Marker-pleasing habits: state the **number**; name **both** the mechanism and the gene/protein; state a **limitation**; write in complete sentences - no bullet fragments; and never, ever answer in Chinese.

Figure E2 - The questions he named, and your 90 minutes

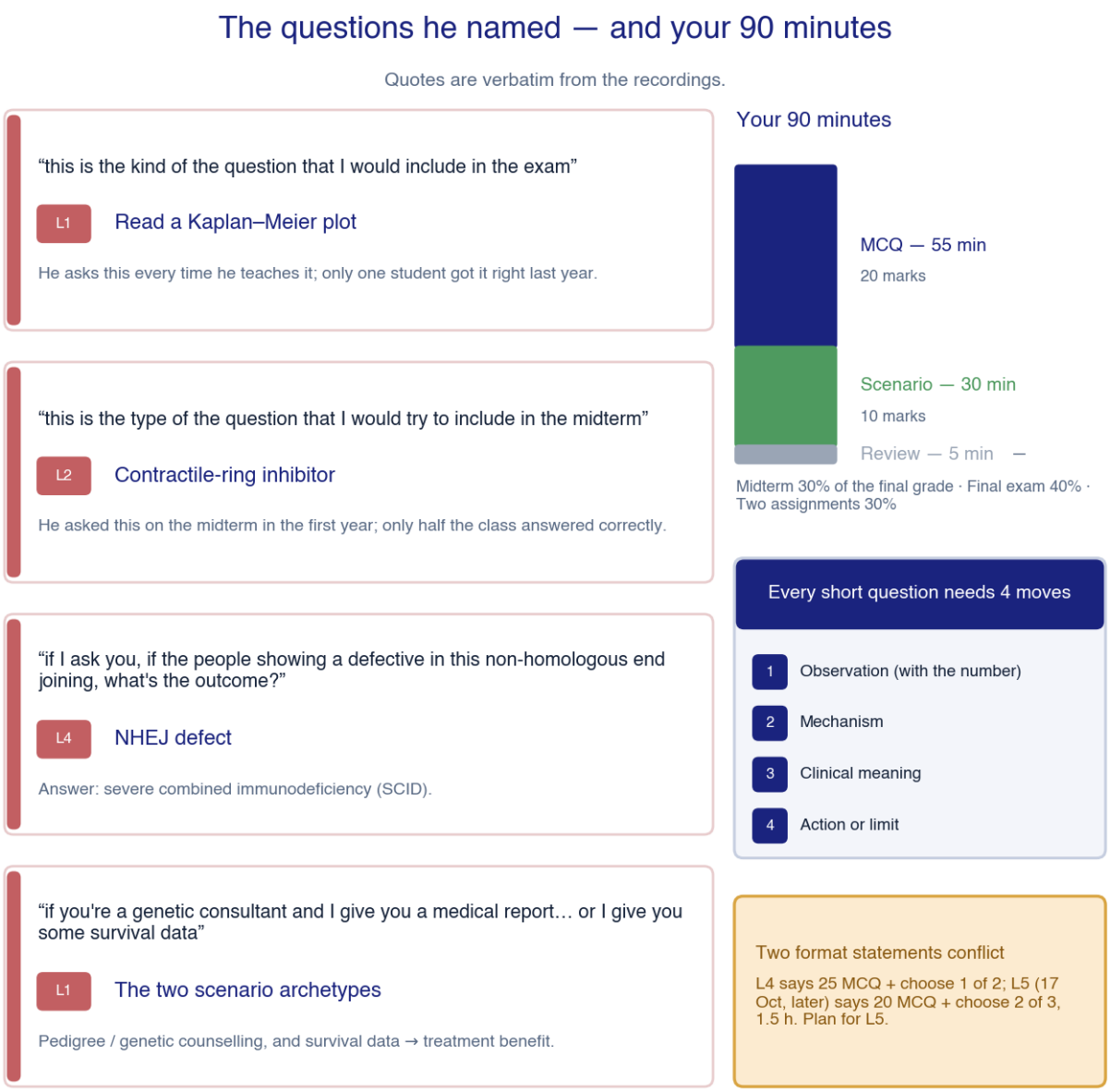


Figure E2. The four named questions with his verbatim quotes, the 90-minute budget, and the four moves every short answer needs. Figure authored at 6.5 in print width and placed 1:1 on this page.

11. How to spend the last 72 hours

Block	Time	Content	Why
A	3 h	Tier 1 3 in full - KM curves, contractile ring, NHEJ->SCID, mitochondrial pedigree, M1/M2	He named these as exam-type
B	2 h	Tier 2 4 - each concept, the discriminating fact, and one clinical consequence	He said "must understand / always emphasize"
C	1.5 h	5 numbers and 7 pairs - drill as flashcards	MCQ fodder
D	2 h	Re-do the 6 highest-yield scenarios from the case bank **in English, in writing, timed**	10 of the 30 marks are scenario
E	1 h	6 exclusions - write a one-page "will NOT be asked" sheet	Stop wasting time
F	1 h	Read your own answers aloud for grammar and structure	Marks are for reasoning, in English

In the exam: MCQs first - they are the passing floor and are pure recall. Then the scenario questions, with the four-move template. Budget roughly **55 minutes for the MCQs, 30 minutes for the short question, 5 minutes to review**.

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12. How confident is this?

High confidence - the exam format, the English-only rule, the four "must understand" statements, the exclusion list, and the Tier 1 questions. These are his own words, captured verbatim from the recordings.

Moderate confidence - Tier 2 and the ordering within Tier 1. A concept he emphasised may still be a Tier 3 MCQ rather than a scenario question.

Lower confidence - the precise split between the two format statements. Lecture 5's "20 MCQ + 2 of 3" and Lecture 4's "25 MCQ + choose 1 of 2" cannot both be right. Lecture 5 is later and is the version written on the slide, so that is the one to plan for. If the paper does not match, the *ratio* and the *strategy* still hold: MCQ for passing, scenarios for the marks.

One honest caveat: this is a prediction from teaching behaviour. No lecturer can be made to state the actual paper, and he said himself that the midterm *"is not that simple"* - so treat this as a prioritisation of your revision, not a guarantee.

Companion documents: `GNBF5020_Complete_Revision_Pack.pdf` *(full notes, English)* - `GNBF5020_中文版溫習手冊.pdf` *(繁體中文)* - `GNBF5020_Case_Study_Bank.pdf` *(16 scenarios with model answers)* - `GNBF5020_Terminology_Deep_Dive.pdf` *(every medical and biology term explained in detail, English)* - `GNBF5020_術語詳解_繁體中文.pdf` *(the same book in Traditional Chinese).*